

A Citation-Aware Legal RAG Pipeline

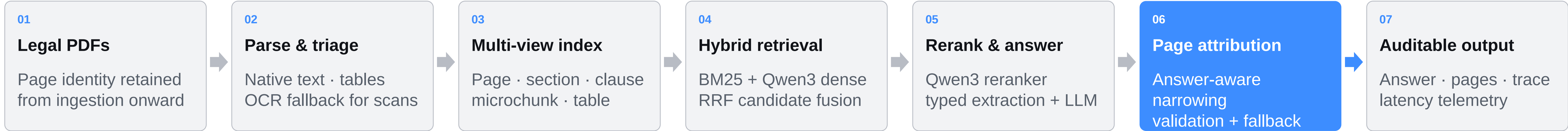
Architecture and page-level grounding evaluation across 100 legal questions

01 HYPOTHESIS

Does answer-aware page attribution improve citation precision without creating complete source misses?

H1 · A separate page-selection layer increases precision while retaining at least one correct page per answer.

02 METHODOLOGY · SYSTEM ARCHITECTURE



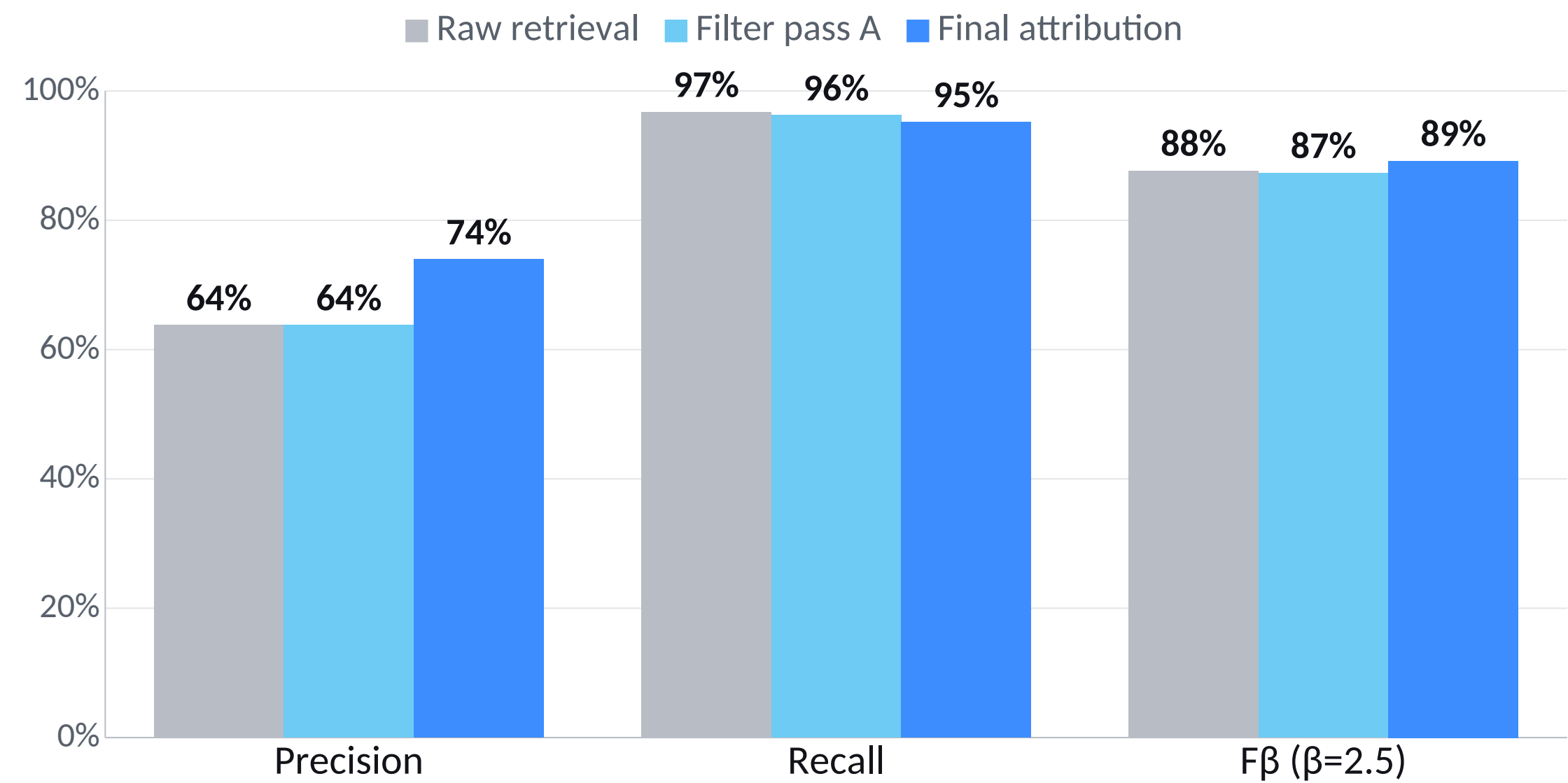
Architectural invariant: every chunk keeps its document_id and page_number, so retrieval, answering, and citations remain traceable.

Experiment

- Corpus:** 30 legal PDFs, 590 pages.
- Questions:** 100 total; 95 have gold page annotations.
- Answer types:** 32 boolean, 30 free text, 17 number, 15 name, 5 names, 1 date.
- Paired comparison:** raw pages → filter pass A → final answer-aware pages.
- Metrics:** macro precision, recall, and F β with $\beta = 2.5$.
- Uncertainty:** 20,000 paired bootstrap resamples over 95 questions.

03 RESULTS · Page-level grounding by stage

Macro averages on 95 questions with annotated source pages



What the result supports

- Observed gain:** final attribution removes excess pages and raises precision from 63.8% to 73.9%.
- Small recall cost:** 96.8% → 95.2%; no question loses every correct page.
- Best fit:** single-page and numeric questions benefit most.
- Failure mode:** multi-page and multi-document questions lose supporting evidence.

Product implication
Keep citation selection explicit; narrow conservatively for multi-page evidence.

Research context

RAG couples retrieval and generation [1]; dense retrieval complements lexical search [2,3]. Citation-aware evaluation treats provenance as a separate quality dimension [4,5].

+10.2 pp

precision
95% CI: +6.3 to +14.5

95 / 95

retain ≥ 1 correct page

−0.33

pages cited per question

Limitations

- One saved run; no repeated seeds. Overall F β : +1.5 pp (95% CI −0.3 to +3.1).
- OCR was not exercised: all 590 pages exposed extractable text.
- Multi-page evidence regresses; mean first-token latency is 6.39 s.

[1] Lewis et al., NeurIPS 2020 · [2] Karpukhin et al., EMNLP 2020 · [3] Cormack et al., SIGIR 2009 · [4] Gao et al., EMNLP 2023 · [5] Pipitone & Alami, 2024 · Model: Qwen3 Embedding/Reranker [6]

Competition context intentionally omitted; evaluation uses the saved legal QA benchmark artifacts.